

Intrusion Detection Using Machine Learning and Deep Learning on CICIoT2023 Dataset

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Abstract—The quick growth of IoT devices has made security risks way higher than before, for instances DDoS, Mirai, reconnaissance and spoofing attacks. In order to study these attacks and risks, this report relies on the CICIoT2023 dataset as a source of the study, and this dataset is a real-world IoT traffic data and this data was collected using more than 105 real physical devices. The dataset also includes 33 different attack types and around 47 features. Recent studies have used ML (machine learning) and DL (deep learning) models on this dataset, the studies achieved high detection accuracy. These approaches including ensemble machine learning and deep neural networks additionally GAN-based augmentation and federated learning and finally transformer models. This report explains very clearly the CICIoT2023 dataset and its features, additionally a review of existing researches that use the dataset for intrusion detections and cyber attacks in Internet of Things (IoT) networks

Keywords—Machine learning, Deep learning, Internet of Things, Cybersecurity, CICIoT2023

I. INTRODUCTION

The Internet of Things or (IoT) in short, connects many devices such as cameras, air conditioners, heat sensors, home assistants and more. All these devices have weak security most of the times, which makes them very easy target to attackers and intruders. Traditional intrusion detection databases such as KDD99 database or even NSL-KDD database were built for normal computer networks and don't actually represent any real IoT traffic. To solve such an issue, CICIoT2023 database was made, and it is a large scale and real world IoT traffic data. It was collected using many real physical devices, the number could reach 105 devices easily. This specific dataset includes real traffic and many types of attacks and intrusions. Because of this, CICIoT2023 dataset has become one of the most important datasets for IoT attacks and intrusion detection research.

II. DATASET DESCRIPTION

The CICIoT2023 dataset is a large scale real world Internet of Things (IoT) traffic dataset. The dataset was collected by using real physical testbeds or devices that can reach the amount of more than 105 IoT devices. These Internet of Thing (IoT) devices like smart cameras, smart washing machines, smart ACs, smart plugs, motion sensors, heat sensors, and smart home hubs. Normal traffic was generated

through real human interactions such as video streaming, voice commands, and device control. As for the attack traffic was generated using a separate attack network. The collected traffic was captured as PCAP files and later converted into CSV format for use in machine learning (ML) and deep learning (DL) models for processing and predicting. The dataset contains 33 different attack types grouped into seven major categories: DDoS, DoS, Mirai botnet, reconnaissance, web-based attacks, brute-force attacks, and spoofing attacks. This allows both binary classification (attack vs normal) and multi-class classification (attack type detection) .

III. DATASET FEATURES

The CICIoT2023 dataset contains around 47 features that describe the behavior of Internet of Thing (IoT) traffic. All the features are extracted from network packets and they are very important for finding the difference between normal traffic and malicious attacks. Each feature represents a statistical or behavioral property of the communication between devices.

One of the main core features is the timestamp, which is the time when each packet is captured. This feature helps in reconstructing and ordering the sequence of events and detecting abnormal traffic speed, such as in DDoS attacks where thousands of packets arrive in a very short time. Another important feature is the flow duration, which represents how long a communication session lasts. Normal Internet of Thing (IoT) communication usually has a specific and predictable durations, while attacks may produce very long or very short flows which off pattern and can be detected easily using that specific feature.

The header length feature describes the size of the packet header. Abnormal header sizes may indicate manipulated packets used in attacks. The protocol type feature tells us what kind of communication protocol it is, such as TCP, UDP, or ICMP. Attackers often abuse specific protocols, for example ICMP is commonly used in ping flood attacks, while UDP is used in amplification attacks.

Traffic speed features such as the rate, source rate (Srate), and destination rate (Drate). These features are representing how fast the packets are being sent and received. High values usually indicate flooding attacks, in the other hand very low values may indicate slow scanning attacks.

Another important features such TCP flag values and flag counts, like SYN, ACK, FIN, RST, PSH, ECE, and CWR. These flags control how connections are created and closed. In normal traffics, flag values are balanced and reasonable. In attack traffics, these values become highly unbalanced, for instance, SYN flood attack produces a very large number of SYN flags without any corresponding ACK flags.

The packet size statistics features describe the average, minimum, maximum, and standard deviation of packet sizes. Normal Internet of Thing (IoT) traffic usually have packet sizes between 200 and 400 bytes. Flooding attacks most of the times use very large packet sizes, on the other hand stealth attacks may use very small packets. The inter arrival time (IAT) feature calculates the time gap between consecutive packets. Very small gaps show high-speed floods, while very large gaps may point to slow attacks such as Slowloris.

Lastly, the dataset includes several statistical balance features such as magnitude, radius, covariance, variance, and weight. These features describe the relationship between the incoming and the outgoing traffic. In normal communication, the incoming and outgoing packets are fairly balanced. In attacks, this won't be the case, and the balance is often broken, for example when a victim receives a very large number of packets but sends very few responses which feels off.

Overall, the mentioned features provide a complete description of Internet of Thing (IoT) traffic behavior in terms of time, speed, protocol usage, packet size and communication balance. These features allow machine learning (ML) and deep learning (DL) models to effectively learn and find the difference between normal Internet of Thing (IoT) behavior and different types of attacks.

#	Feature	Description
1	ts	Timestamp
2	flow duration	Duration of the packet's flow
3	Header Length	Header Length
4	Protocol Type	IP, UDP, TCP, ICMP, Unknown (Integers)
5	Duration	Time-to-Live (ttl)
6	Rate	Rate of packet transmission in a flow
7	Srate	Rate of outbound packets transmission in a flow
8	Drate	Rate of inbound packets transmission in a flow
9	fin flag number	Fin flag value
10	syn flag number	Syn flag value
11	rst flag number	Rst flag value
12	psh flag numbe	Psh flag value
13	ack flag number	Ack flag value
14	ece flag numbe	Ece flag value
15	cwr flag number	Cwr flag value
16	ack count	Number of packets with ack flag set in the same flow
17	syn count	Number of packets with syn flag set in the same flow
18	fin count	Number of packets with fin flag set in the same flow
19	urg coun	Number of packets with urg flag set in the same flow
20	rst count	Number of packets with rst flag set in the same flow
21	HTTP	Indicates if the application layer protocol is HTTP
22	HTTPS	Indicates if the application layer protocol is HTTPS
23	DNS	Indicates if the application layer protocol is DNS
24	Telnet	Indicates if the application layer protocol is Telnet
25	SMTP	Indicates if the application layer protocol is SMTP
26	SSH	Indicates if the application layer protocol is SSH
27	IRC	Indicates if the application layer protocol is IRC
28	TCP	Indicates if the transport layer protocol is TCP
29	UDP	Indicates if the transport layer protocol is UDP
30	DHCP	Indicates if the application layer protocol is DHCP
31	ARP	Indicates if the link layer protocol is ARP
32	ICMP	Indicates if the network layer protocol is ICMP
33	IPv	Indicates if the network layer protocol is IP
34	LLC	Indicates if the link layer protocol is LLC
35	Tot sum	Summation of packets lengths in flow
36	Min	Minimum packet length in the flow
37	Max	Maximum packet length in the flow
38	AVG	Average packet length in the flow
39	Std	Standard deviation of packet length in the flow
40	Tot size	Packet's length
41	IAT	The time difference with the previous packet
42	Number	The number of packets in the flow
43	Magnitue	(Average of the lengths of incoming packets in the flow + Average of the lengths of outgoing packets in the flow) ** 0.5
44	Radius	(Variance of the lengths of incoming packets in the flow + Variance of the lengths of outgoing packets in the flow) ** 0.5
45	Covariance	Covariance of the lengths of incoming and outgoing packets
46	Variance	Variance of the lengths of incoming packets in the flow / The variance of the lengths of outgoing packets in the flow
47	Weight	Number of incoming packets * Number of outgoing packets

IV. LITERATURE REVIEW

Researchers from around the world have turned to the CICIOT2023 dataset to better understand how to detect intrusions in IoT networks. They've explored a variety of approaches, including traditional machine learning, deep learning, ensemble learning, GAN-based data augmentation, federated learning, and even transformer models, all in an effort to find more effective solutions [1], [2].

A. Traditional Machine Learning Approaches

Some researches are focusing on using traditional machine learning (ML) models for attack and intrusion detections. One of the researches used an ensemble of Naive Bayes and Support Vector machine classifiers to detect reconnaissance attacks in Internet of thing (IoT) networks. The dataset was preprocessed firstly by handling the missing values by deleting them and then normalizing features, encoding categorical values, and applying data splitting. This model achieved about 99% accuracy with very high precision, recall, F1-Score, and AUROC. But this method is mainly focusing on binary classification and didn't perform full multi-class attack detection [1].

Another study did compare multiple machine learning (ML) algorithms, like Logistic regression, Naive Bayes, Support vector machine, Random Forest, Decision Tree, and K-nearest neighbor. The results showed that decision tree and random forest achieved the highest results with around 99% accuracy. As for Logistic regression, it gave lower accuracy and that's because it couldn't model a complex IoT attack behavior well. All these researches proved that lightweight models can be effective for Internet of Thing (IoT) intrusion detection [3], [4].

B. Deep Learning Based Detection

Deep learning was applied to the CICIOT2023 dataset. Many researches evaluated deep models such as CNN, LSTM, GRU, Autoencoders, and CNN-LSTM hybrid networks for multi-class Internet of Thing (IoT) attack and intrusion detection. These models were trained after applying preprocessing steps such as MinMax normalization, one hot encoding, and data splitting. The CNN-LSTM model (Hybrid model) did achieve an accuracy between 97% and 99%, which was the best performance [5].

Another deep learning system proposed a real time intrusion detection and prevention system using a deep multilayer perceptron or as a known shortcut (MLP) with more than 10 layers and ReLU activation functions. This system was trained on CICIOT2023 dataset and integrated with iptables for real-time attack blocking. The model did achieve more than 85% overall accuracy and very high precision for Mirai attacks. Although the model performed well, it has high hardware requirements and some heavy preprocessing because the dataset was too large [2].

C. Gan and Federated Learning Based Solutions

One advanced solution combined Generative Adversarial Networks (GANs) with Federated Learning (FL) to make intrusion detection performance better, stronger, more precise, and solve data imbalance and privacy problems. (GANs) were used to generate samples for minority attack classes, while federated learning was used to train the model in a decentralized way. XGBoost was used as the core classifier. Without GAN augmentation, the model achieved

only about 67% accuracy. With GAN-generated data, the accuracy increased to more than 95% which improves how it affect the result. Its true that this solution provided strong and good performance, but as a trade-off, it increased system complexity and training cost [6].

D. Transformer Based Intrusion Detection

Transformer based deep learning model was also applied to the dataset (CICIoT2023 dataset) for multi-class intrusion or attack detection. The researchers compared many deep learning (DL) models like DNN, CNN, RNN, LSTM, and finally CNN based hybrids with customized transformer encoders. The results did show that it's true that the transformer wasn't the best model for binary classification It did somehow achieve around 99% accuracy for multi-class attack classifications. This tells us that transformer models can effectively learn very complex patterns in IoT traffic [7].

E. Data Leakage and Feature Selection Impact

An important research analyzed the effect of data leakage during feature selection while using the dataset (CICIoT2023 dataset). Researches proved that selection features before splitting the dataset actually lead to unreal high accuracies, even higher than 99%. When the correct pipeline or steps was followed and applied (split first then scale and select features), the accuracy went lower for some of the models, especially deep learning (DL) models like for instance RNN model. This research showed the importance of correct preprocessing steps to avoid misleading results [8].

F. Summary of Existing Research

From all these researches, it is clear that the CICIoT2023 dataset is very effective for both machine learning (ML) and deep learning (DL) for intrusion detection. Traditional machine learning (ML) models provide high accuracy with low computational cost [1], [9], [3] on the other hand deep learning (DL) and advanced methods such as GANs, federated learning, and transformers gives stronger and better detection performance but as a tradeoff they are costly, more complex and use a lot of resources [6], [5], [10], [7], [2].

V. CONCLUSION

This report presented an overview of the dataset (CICIoT2023) and its use in Internet of Thing (IoT) attack or intrusion detection. The dataset has real world traffic collected from more than 105 real physical devices and contains around 33 different cyber-attack types with 47 network features. The reviewed research show how both machine learning (ML) and deep learning (DL) models achieve high accuracy when detecting attacks or intrusions using the CICIoT2023 dataset. On the other hand, challenges like data imbalance, high computational cost and correct preprocessing steps remain an important aspect for future research.

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REFERENCES

- [1] A. Sharma, H. Babbar, and A. K. Vats, "Detecting Reconnaissance Activities in IoT Networks Using UNB CIC-IOT 2023 Dataset," in *2024 4th Asian Conference on Innovation in Technology, ASIANCON 2024*, Institute of Electrical and Electronics Engineers Inc., 2024. doi: 10.1109/ASIANCON62057.2024.10837951.
- [2] B. Taşçı, "Deep-Learning-Based Approach for IoT Attack and Malware Detection," *Applied Sciences (Switzerland)*, vol. 14, no. 18, Sep. 2024, doi: 10.3390/app14188505.
- [3] S. R. Alve, M. Z. Mahmud, S. Islam, Md. A. Chowdhury, and J. Islam, "Smart IoT Security: Lightweight Machine Learning Techniques for Multi-Class Attack Detection in IoT Networks," p. 31, Jul. 2025, doi: 10.1109/QPAIN66474.2025.11171769.
- [4] S. Singh, A. Singh, and V. Goyal, "An Ensemble Machine Learning Approach for Denial-of-Service Detection in Cloud-of-Things Networks Using CICIoT2023 Dataset," 2025. [Online]. Available: <https://internationalpubs.com>
- [5] S. Abbas *et al.*, "Evaluating deep learning variants for cyber-attacks detection and multi-class classification in IoT networks," *PeerJ Comput Sci*, vol. 10, 2024, doi: 10.7717/peerj-cs.1793.
- [6] S. Alahmari and N. Aleisa, "Enhancing the CIC IoT Dataset 2023 for Improved Attack Detection through GANs Augmentation and Federated Learning," *Journal of Computer Science*, vol. 21, no. 7, pp. 1688–1704, Jul. 2025, doi: 10.3844/JCSSP.2025.1688.1704.
- [7] S. M. Tseng, Y. Q. Wang, and Y. C. Wang, "Multi-Class Intrusion Detection Based on Transformer for IoT Networks Using CIC-IoT-2023 Dataset," *Future Internet*, vol. 16, no. 8, Aug. 2024, doi: 10.3390/fi16080284.
- [8] S. B. Mallampati and S. Hari, "A Comparative Study on the Impacts of Data Leakage During Feature Selection using the CIC-IoT 2023 Intrusion Detection Dataset," in *10th International Conference on Electrical Energy Systems, ICEES 2024*, Institute of Electrical and Electronics Engineers Inc., 2024. doi: 10.1109/ICEES61253.2024.10776873.
- [9] A. I. Jony and A. K. B. Arnob, "Securing the Internet of Things: Evaluating Machine Learning Algorithms for Detecting IoT Cyberattacks Using CIC-IoT2023 Dataset," *International Journal of Information Technology and Computer Science*, vol. 16, no. 4, pp. 56–65, Aug. 2024, doi: 10.5815/ijitcs.2024.04.04.
- [10] S. K. Erskine, "Real-Time Large-Scale Intrusion Detection and Prevention System (IDPS) CICIoT Dataset Traffic Assessment Based on Deep Learning," *Applied System Innovation 2025, Vol. 8, Page 52*, vol. 8, no. 2, p. 52, Apr. 2025, doi: 10.3390/ASI8020052.